Spatial–Temporal–Spectral Unified Modeling for Remote Sensing Dense Prediction

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***Abstract*—Despite advances in deep learning for remote sensing dense prediction, a unified framework for handling heterogeneous data, multiple tasks, and diverse semantic category sets remains elusive. Concretely, multiple correlated dense prediction tasks (e.g., semantic segmentation and change detection) are often modeled independently, requiring separate networks or task-specific decoder heads. Moreover, the semantic label space is typically fixed during training, so changing the target class subset requires modifying and retraining the prediction head. Here, we introduce the spatial–temporal–spectral unified network (STSUN), a unified framework for remote sensing dense prediction. The model employs metadata-informed unified representations generated via hypernetworks, enabling adaptation to inputs with variable spatial resolutions, temporal lengths, and spectral band configurations. Furthermore, it unifies diverse dense prediction tasks within a single architecture through learnable task embeddings and semantic category embeddings that support multiple class subsets represented during training. Experiments under scenario-level joint training on datasets with diverse spatial–temporal–spectral configurations demonstrate the effectiveness of the proposed method for modeling multiple dense prediction tasks and semantic class subsets within a single model.**

***Index Terms*—Category unification, data unification, dense prediction, dimensional unification, task unification.**

I. Introduction

THE rapid advancement of remote sensing technologies and the diversification of data acquisition methods have propelled the field into an era of unprecedented growth [1], [2]. Large-scale, multisource remote sensing data are now extensively utilized in various dense prediction tasks, playing a pivotal role in applications such as urban expansion monitoring [3], land cover classification [4], and disaster damage assessment [5]. In dense prediction, remote sensing data, task definitions, and semantic category sets exhibit high heterogeneity and diversity [6]. Remote sensing data are inherently diverse in modality and spatiotemporal scale, encompassing modalities such as RGB, multispectral, hyperspectral, and synthetic aperture radar (SAR). It spans spatial scales from individual buildings to expansive urban areas, and temporal scales ranging from a single day to several decades. Dense prediction tasks primarily include semantic segmentation, binary change (BC) detection, and semantic change (SC) detection [2]; while each task has its distinct objective, intrinsic correlations exist among them. Furthermore, semantic category sets vary significantly across different remote sensing application scenarios, with each scenario possessing a unique set of classes. For instance, an urban scenario may encompass categories such as {buildings, roads, playgrounds, and vehicles}, whereas a forest scenario might include {pine forests, bamboo forests, tea plantations, and shrubs}.

Despite the remarkable progress of deep learning in remote sensing dense prediction, existing architectures are typically tailored to specific data–task–category settings [7], [8], [9]. A variation in the remote sensing data, dense prediction task, or semantic category set commonly requires architectural modification, retraining, or fine-tuning [10]. This limitation prevents current architectures from fully leveraging diverse dense prediction data or learning the latent relationships among different tasks. Requiring a distinct model for each specific data–task–category configuration results in low efficiency and hinders adaptability to diverse dense prediction environments. In particular, the limitations of existing architectures are threefold.

1) *Fixed Data:* Current models are generally designed for specific data modalities and spatiotemporal scales, struggling to accommodate diverse input formats. For example, the change detection model FCCDN [10] is configured for bitemporal RGB inputs. Changing the input format generally requires dataset-specific adaptation.
2) *Fixed Tasks:* For varying dense prediction tasks, existing architectures often employ isolated models or task-specific decoding heads for independent modeling. They

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Data is available online at https://github.com/NJU-LHRS/Official_TSSUN.
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are either restricted to a single task or rely on paired annotations across multiple tasks, limiting the exploitation of correlations among them. For instance, BiT [9] is designed for BC detection, while SIINet [11] is designed for semantic segmentation.

3) *Fixed Category Sets:* Existing supervised architectures usually hard-code a dataset-specific output space and cannot switch among different label subsets through a shared prediction mechanism. For instance, because the ISPRS Potsdam [12] and DeepGlobe [13] datasets use different semantic taxonomies, conventional semantic segmentation methods such as SAPNet [14] require separate output heads and training for each dataset.

To overcome the limitations of current dense prediction models, we conduct a formalized analysis of dense prediction tasks. Remote sensing dense prediction aims to assign pixelwise labels to Earth observation imagery [2], primarily encompassing semantic segmentation, BC detection, and SC detection. Formally, let the dense prediction model be defined as a mapping function $\mathcal{F} : \mathbf{X} \to \mathbf{Y}$, where the input $\mathbf{X}$ and output $\mathbf{Y}$ are tensors defined as follows:

$$\mathbf{X} \in \mathbb{R}^{T_1 \times C_1 \times H_1 \times W_1}, \quad \mathbf{Y} \in \mathbb{R}^{T_2 \times C_2 \times H_2 \times W_2} \tag{1}$$

where T, C, H, and W denote the temporal length, channel (or class) count, height, and width, respectively. Assuming spatial consistency, $(H_1, W_1) = (H_2, W_2) = (H, W)$. Consequently, the specific dense prediction task is determined by the constraints imposed on the input–output temporal (T_1, T_2) and spectral (C_2) relationships

$$\begin{cases} \text{Semantic Segmentation:} & T_2 = 1, \quad C_2 \geq 2 \\ \text{Binary Change Detection:} & T_2 = T_1 - 1, \quad C_2 = 2 \\ \text{Semantic Change Detection:} & T_2 = T_1, \quad C_2 \geq 2. \end{cases} \tag{2}$$

Physically, the input dimensions T_1, C_1, and (H_1, W_1) explicitly parameterize the temporal coverage, data modality, and spatial resolution, respectively. The output dimensions T_2 and C_2 correspond to the task type and semantic category set.

This in-depth analysis reveals that the reliance of existing models on fixed data–task–category settings is fundamentally tied to rigid input and output spatial–temporal–spectral dimensions. Models restricted to a fixed image size (H_1, W_1), temporal length T_1, or spectral band count C_1 are inevitably bound to specific remote sensing data. Similarly, constraining a model to output a fixed temporal length T_2 for a given input length T_1 limits it to a single dense prediction task. Furthermore, mandating a fixed output channel count C_2 confines the model to predicting a predefined semantic category set. This intrinsic relationship between tensor dimensions and the data–task–category paradigm underscores the critical necessity for a unified modeling approach across spatial–temporal–spectral dimensions.

Motivated by the aforementioned analysis, we propose the spatial–temporal–spectral unified network (STSUN), as illustrated in Fig. 1. By unifying the spatial, temporal, and spectral dimensions of inputs and outputs, STSUN accommodates varying image sizes, temporal lengths, and spectral bands within each jointly trained scenario. It executes multiple dense prediction tasks and selects among the class subsets represented during training. STSUN achieves this unification through two core mechanisms.

1) *Dimension Unification Strategy (DUS):* Unlike a coupled strategy that directly unifies all spatial–temporal–spectral dimensions at the input and output, we use a targeted, progressive sequence. First, unifying the input spatial–spectral dimensions enables the model to extract the spatial–spectral features of each temporal phase. Subsequently, unifying the input and output temporal dimensions performs feature-level fusion of multitemporal representations, reducing the interference associated with early pixel-level fusion. Because the temporal dimension also specifies the dense prediction task type, its independent unification and adaptive adjustment help the shared model represent multiple tasks. Finally, unifying the output spatial–spectral dimensions adjusts the output grid and class channels according to the target metadata.
2) *Dimension Unification Module (DUM):* To implement unification across all dimensions, we design DUM. The metadata of each dimension characterizes its configuration; for instance, spectral wavelengths describe the input channels, while the category metadata identifies the classes corresponding to output channels. DUM feeds this metadata into a hypernetwork that generates an adaptive mapping network. The generated mapping projects a variable input into a unified representation along the selected dimension or projects the unified representation into a requested output configuration.

To validate the data–task–category unification capability of STSUN in dense prediction, we select building and land use/land cover (LULC) dense prediction as representative scenarios, conducting experiments on four building datasets and three LULC datasets. These datasets feature diverse data–task–category configurations: image sizes range from 256×256 to 1024×1024, spatial resolutions from 0.075 to 10 m, temporal lengths from 1 to 24, temporal sampling intervals from 1 month to 14 years, and input spectral configurations from 3 to 10 bands. Furthermore, the evaluated tasks encompass semantic segmentation, BC detection, and SC detection, with category sets varying from 2 to 7 classes.

In summary, the primary contributions of this work are threefold.

1) We propose STSUN, which unifies the spatial, temporal, and spectral dimensions of dense-prediction inputs and outputs. After scenario-level joint training, one model can switch among the data configurations, tasks, and class subsets represented in that scenario without dataset-specific heads; this setting does not imply zero-shot generalization to unseen classes.
2) We design the DUS and the DUM. DUS instantiates a feature-level fusion schedule within the variable-input/variable-output framework, while DUM uses dimensional metadata and hypernetworks to adaptively map heterogeneous data formats to and from a shared representation.

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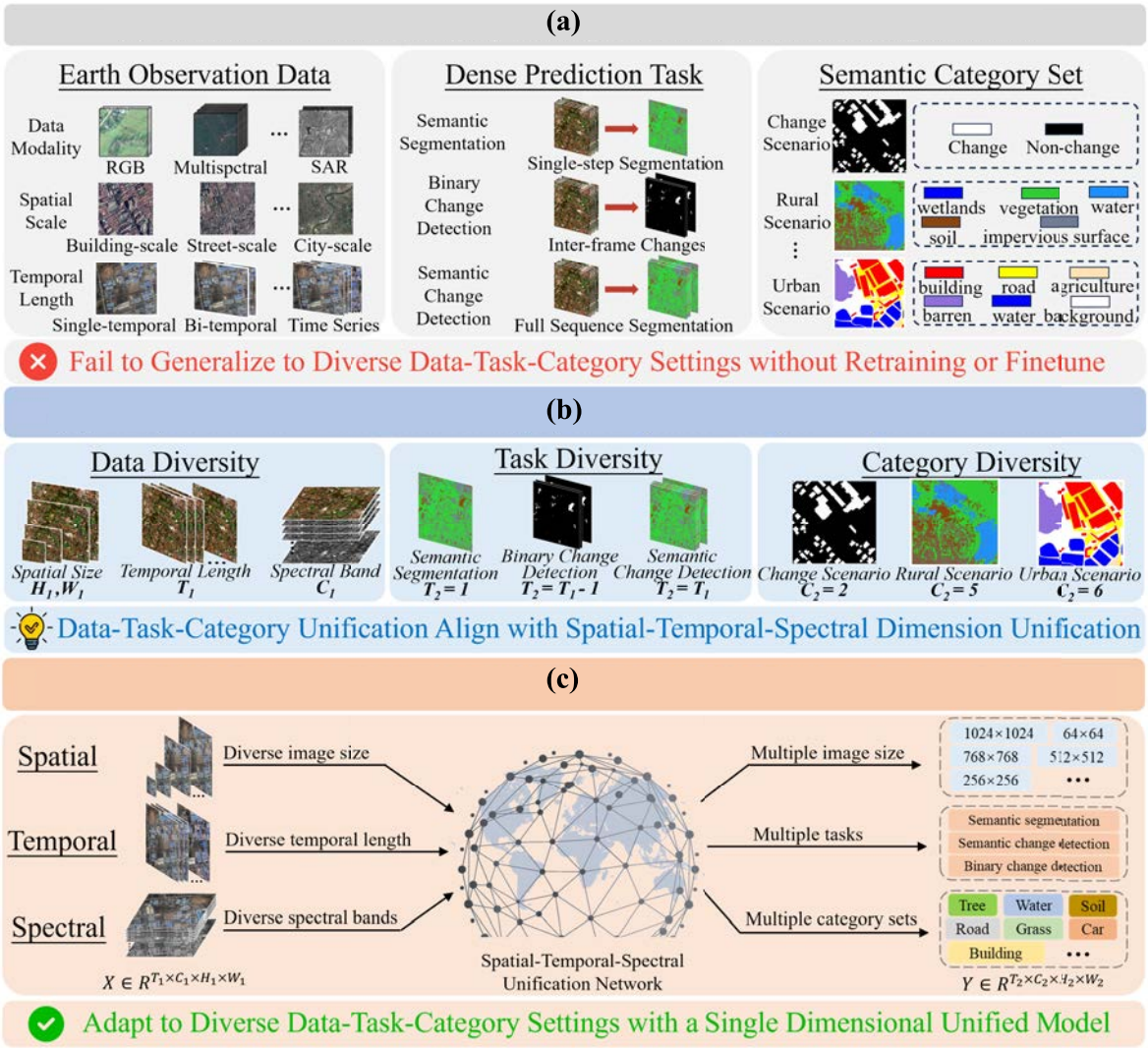


Fig. 1. Illustration of the STSUN. Within a scenario-level jointly trained model, STSUN handles multiple spatial, temporal, and spectral input–output configurations and unifies semantic segmentation, BC detection, and SC detection with selectable category subsets. (a) diverse data–task–category setting and limitations of existing methods. (b) Analysis between data–task–category and spatial–temporal–spectral dimensions. (c) Data–task–category unification with dimensional unification strategy and module.

3) Experiments on building and LULC datasets validate the data–task–category unification capability of STSUN across varying image sizes, temporal lengths, spectral bands, dense prediction tasks, and category subsets. The results show competitive performance while using one jointly trained model per scenario.

II. Related Works

A. Remote Sensing Data Heterogeneity

Remote sensing data exhibits significant heterogeneity across spatial, temporal, and spectral dimensions [15]. Spatially, differences in ground sampling distance and image extent hinder generalization, as models trained at a given resolution or image size often perform poorly at others due to changes in the scale and contextual appearance of ground objects [16]. Temporally, data availability ranges from bitemporal pairs used for BC detection to dense time series with hundreds of observations. This disparity in sequence length makes models trained on one temporal length incompatible with others. Spectrally, the number and type of bands vary significantly between sensors, from single panchromatic bands to hundreds of hyperspectral bands [17]. As a result, models are often tied to specific sensors. While some approaches attempt to unify inputs from different sensors using techniques like hypernetworks, they often overlook other dimensions [18], [19], [20]. The core research gap is a unified model architecture that can ingest and process this diverse spatial–temporal–spectral data at both the input and output levels.

B. Dense Prediction Task Unification

While dense prediction encompasses tasks including semantic segmentation, BC detection, and SC detection [21], current unification efforts are limited. Multitask learning frameworks have shown promise, leveraging shared representations to improve performance and efficiency [22]. Recent methods in remote sensing have increasingly adopted multitask learning

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to address multiple dense prediction tasks within a unified framework [23]. However, these methods suffer from two major drawbacks. First, they exhibit a strong reliance on annotations of all tasks, which limits the pool of usable training data [21]. Second, they often rely on task-specific decoders, which prevents the full exploitation of synergies between tasks during feature synthesis [10]. Consequently, a significant research gap exists for a unified model that can jointly learn multiple tasks from unpaired datasets.

C. Flexible Semantic Class Sets

The semantic categories required for remote sensing tasks are highly dependent on specific applications and geographic scenes, ranging from simple binary sets to complex multiclass taxonomies. However, SOTA models [7], [8], [9] are typically designed for fixed class sets, where the classification layer is hard-coded to a specific number of outputs. This rigidity limits their generalization across diverse tasks [24]. Recent remote sensing open vocabulary semantic segmentation (OVSS) methods, such as OVRS [25] and SegEarth-OV [26], address a related but distinct objective: recognizing categories beyond the segmentation label space by transferring language-aligned semantic knowledge from vision–language or foundation models, with target categories specified through textual names or prompts. STSUN instead considers a closed-set multitaxonomy problem. All candidate semantic classes belong to a predefined library of trainable category embeddings learned solely from visually annotated remote sensing data, while each dataset or application scenario selects the subset required by its taxonomy.

III. Methodology

A. Problem Formulation

Dense prediction in remote sensing encompasses a range of tasks that, despite their distinct objectives, share a common foundation: inferring structured, pixelwise semantic information from multidimensional earth observation data. We formalize these tasks within a unified mathematical framework by conceptualizing them as a highly flexible tensor-to-tensor mapping problem.

Let an input remote sensing data instance be represented by a primary data tensor $\mathbf{X} \in \mathbb{R}^{T_1 \times C_1 \times H_1 \times W_1}$, while the corresponding model output is a prediction tensor $\mathbf{Y} \in \mathbb{R}^{T_2 \times C_2 \times H_2 \times W_2}$, consistent with the formulation in Section I.

To address variability in data–task–category-specific requirements, we augment this core tensor representation with explicit metadata. The model's behavior is conditioned not only on $\mathbf{X}$ but also on two metadata sets: an input set $\mathbf{M}_1$ describing the source data and an output set $\mathbf{M}_2$ specifying the target prediction structure. These sets are decomposed as follows.

1) *Input Metadata:* $\mathbf{M}_1 = \{\mathbf{M}_{(H_1,W_1)}, \mathbf{M}_{T_1}, \mathbf{M}_{C_1}\}$. This set provides context about $\mathbf{X}$. $\mathbf{M}_{(H_1,W_1)}$ encodes spatial information, including pixel locations and spatial resolution. $\mathbf{M}_{T_1}$ contains the acquisition timestamps for the T_1 temporal slices. $\mathbf{M}_{C_1}$ specifies the wavelengths of the C_1 spectral channels.
2) *Output Metadata:* $\mathbf{M}_2 = \{\mathbf{M}_{(H_2,W_2)}, \mathbf{M}_{T_2}, \mathbf{M}_{C_2}\}$. This set specifies the desired output format and task. $\mathbf{M}_{(H_2,W_2)}$ defines the pixel locations and spatial resolution of the target prediction grid $\mathbf{Y}$. $\mathbf{M}_{T_2}$ contains the target temporal length T_2 and a trainable task embedding identifying the requested dense prediction task. $\mathbf{M}_{C_2}$ defines the prediction's semantic space as a selected subset of trainable class embeddings corresponding to the C_2 target classes.

Within this framework, our goal is to learn a single, unified mapping function f_θ parameterized by θ, capable of adapting to multiple input and output configurations. The general mapping is defined as follows:

$$\mathbf{Y} = f_\theta(\mathbf{X}, \mathbf{M}_1, \mathbf{M}_2). \tag{3}$$

The three core tasks, including semantic segmentation, BC detection, and SC detection, as well as the category subsets represented during training, are handled as instances of this general function. For each scenario, the model maintains a task-embedding library indexed by task identity and a class-embedding library indexed by semantic class identity. The metadata of each training sample selects one task embedding and the relevant ordered subset of class embeddings. These selected embeddings and the shared hypernetwork receive gradients jointly with the backbone; embeddings not selected for that sample receive no direct gradient from its loss. Consequently, datasets that request the same task or semantic class share the corresponding embedding rather than allocating a dataset-specific decoder. Adding a represented task or class expands only the relevant embedding library linearly, while the encoder, decoder, and hypernetwork remain shared. This is a supervised closed-library mechanism and does not provide zero-shot prediction for an unseen class.

B. Overview of the STSUN

STSUN achieves unified data representation across spatial, temporal, and spectral dimensions by leveraging input and output metadata. As illustrated in Fig. 2, the STSUN architecture comprises five distinct stages, utilizing trainable task and class embeddings to flexibly adapt to diverse input/output configurations and target category sets.

First, the input spatial–spectral unified module (ISSUM) unifies the spatial and spectral dimensions. By exploiting the inherent continuity of spectral signatures and spatial proximity, ISSUM employs hypernetworks conditioned on $M_{(H_1,W_1)}$ and M_{C_1} to generate adaptive linear layers. These layers map the heterogeneous input into a unified representation while strictly maintaining temporal independence to prevent degradation during multitemporal fusion. Notably, spatial unification is not intended merely to accommodate different numbers of spatial tokens, which attention mechanisms can already handle. Rather, it addresses the heterogeneous physical spatial scales represented by different image grids. By conditioning the spatial mapping on image size, pixel location, and spatial resolution, ISSUM adaptively projects spatial configurations with different ground extents and object scales into a shared representation before attention-based feature extraction.

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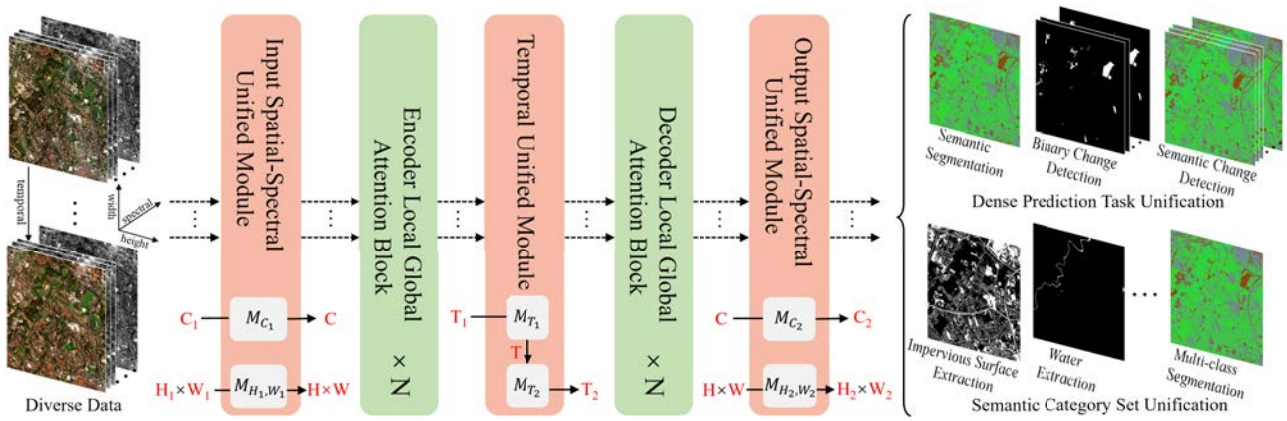


Fig. 2. Overview of the STSUN architecture. The highlighted shapes on either side of each unified module indicate its unified dimension.

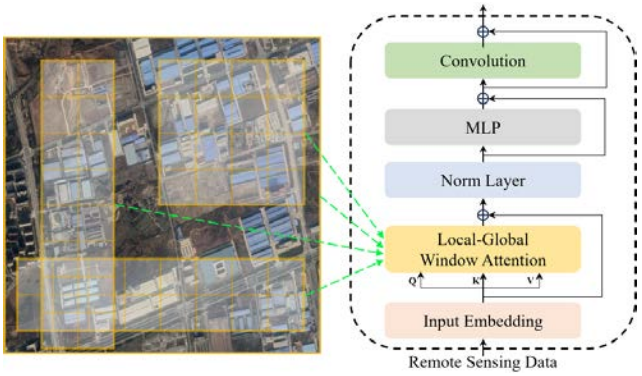


Fig. 3. Architecture of the LGWA block. Three local-attention branches with square, horizontal, and vertical windows operate in parallel. Their fused representation is then processed by global attention in series.

Second, to ensure robust feature extraction across varied remote sensing scenarios, a shared-weight encoder processes each temporal image independently. This encoder is constructed from local–global window attention (LGWA) blocks (Fig. 3). Each block applies three local-attention branches in parallel using square, horizontally elongated, and vertically elongated windows. Their outputs are fused before a global-attention operation is applied serially, so local geometry is captured before scene-level dependencies are modeled. The same branch topology is used at the corresponding encoder and decoder stages.

Third, the temporal unified module (TUM) bridges the encoder and decoder by unifying the temporal dimension. Leveraging the continuity of temporal changes, TUM uses a hypernetwork conditioned on $\mathbf{M}_{T_1}$ to generate adaptive parameters for feature-level fusion across temporal phases. Subsequently, a trainable task embedding guides the network toward the requested dense prediction task. In particular, $\mathbf{M}_{T_2}$ conditions a hypernetwork that maps the unified representation to the required output temporal length, thereby supporting the joint modeling of multiple tasks.

Fourth, a shared-weight decoder refines the fused, task-guided features. Notably, the decoder employs the same LGWA modules as the encoder. By applying this symmetrical local–global attention mechanism, the decoder effectively suppresses irrelevant information and precisely aligns the refined representations with the specific semantic requirements of the target task.

Fifth, the output spatial–spectral unified module (OSSUM) finalizes the spatial and spectral mapping. It restores features to the target spatial grid using an adaptive layer generated from $\mathbf{M}_{(H_2,W_2)}$ and subsequently maps them to class-specific predictions using a layer conditioned on $\mathbf{M}_{C_2}$. The metadata $\mathbf{M}_{C_2}$ selects the target category subset from the trained class-embedding library.

By structuring the processing pipeline across these five stages, STSUN effectively accommodates highly diverse data configurations, enabling the high-performance joint modeling of multiple tasks and category sets.

C. Dimension Unification Module

To facilitate the mapping from the original data space to the unified feature space across different dimensions (spatial, temporal, and spectral), and to transform the unified features into outputs of appropriate shapes according to the specific task and category requirements, STSUN introduces the DUM. This module serves as the fundamental implementation of each dimension unification in ISSUM, TUM, and OSSUM. DUM employs dimensional metadata and trainable embeddings to generate adaptive weights and biases for linear layers, thereby enabling adaptive and metadata-aware feature transformation.

Given an input tensor $\mathbf{X} \in \mathbb{R}^{L\times D_1}$ and its metadata $\mathbf{M} \in \mathbb{R}^{D_1}$, as well as trainable embeddings $\mathbf{E} \in \mathbb{R}^{D_1}$ when applied to output temporal and spectral dimensions, which indicates the task or semantic class set, DUM maps the varying input tensor $\mathbf{X}$ to a unified feature $\mathbf{Y} \in \mathbb{R}^{L\times D_2}$, or maps a unified feature $\mathbf{X}$ into the varying output $\mathbf{Y}$, as illustrated in Fig. 4. D_1 and D_2 can refer to spatial, temporal, or spectral dimensions. The detailed procedure is as follows.

1) *Metadata Embedding:* Metadata $\mathbf{M}$ is tokenized by a linear layer, and the positional encoding and optional trainable embeddings $\mathbf{E}$ are added to it. A learnable class token [CLS] is prepended to the token sequence $\mathbf{T} \in \mathbb{R}^{(D_1+1)\times D_2}$.
2) *Cross-Variable Relationship Modeling:* The metadata token sequence $\mathbf{T}$ is processed through multiple transformer blocks to capture the underlying relationships among metadata.

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TABLE I

BRIEF INTRODUCTION OF THE EXPERIMENTAL DATASETS. LULC DENOTES LAND USE/LAND COVER. SS, BCD, AND SCD DENOTE SEMANTIC SEGMENTATION, BINARY CHANGE DETECTION, AND SEMANTIC CHANGE DETECTION, RESPECTIVELY

Name	Scenario	Task	T_1	T_2	C_1	C_2	H, W	Resolution (m)	Samples	Interval
WHU [27]	Building	SS	1	1	3	2	512×512	0.3	8189	N/A
WHU-CD [27]	Building	SCD	2	2	3	2	1024×1024	0.075	480	4 years
LEVIR-CD [28]	Building	BCD	2	1	3	2	1024×1024	0.5	637	5–14 years
TSCD [29]	Building	BCD	4	3	3	2	256×256	0.5	2700	2 years
LoveDA [30]	LULC	SS	1	1	3	7	1024×1024	0.3	5987	N/A
DynamicEarthNet [31]	LULC	BCD & SCD	24	23 & 24	4	2 & 6	1024×1024	3	54750	1 month
MultiSenGE [32]	LULC	SS	1	1	10	5	256×256	10	8157	N/A

3) *Adaptive Linear Layer Generation:* T is divided into the [CLS] token and other tokens. The [CLS] token is linearly projected to generate bias parameters $\mathbf{b} \in \mathbb{R}^{D_2}$, while the remaining tokens are projected to form a weight matrix $\mathbf{W} \in \mathbb{R}^{D_1 \times D_2}$. The resulting $\mathbf{W}$ and $\mathbf{b}$ form a linear layer that maps the input features with D_1 channels to output features with D_2 channels.
4) *Adaptive Feature Mapping:* The input $\mathbf{X} \in \mathbb{R}^{L \times D_1}$ can be processed through the generated linear layer, resulting in an output $\mathbf{Y} \in \mathbb{R}^{L \times D_2}$.

The effectiveness of DUM in unifying dimensions can be exemplified by the input spectral and spatial dimensions. For spectral unification, multispectral and hyperspectral remote sensing data contain varying numbers of spectral channels, each associated with a specific wavelength. Due to the continuity of spectral wavelengths, these datasets can be interpreted as different sampling rates in the spectral domain. DUM leverages this continuity to effectively encode diverse spectral inputs within a unified representation. For spatial unification, the spatial positions of the feature map are organized as the variable dimension D_1, while the remaining feature dimensions are absorbed into L. The spatial metadata are processed by the hypernetwork to generate $\mathbf{W}$ and $\mathbf{b}$ for the current spatial configuration, and the resulting linear mapping is consistently applied along the remaining feature dimensions. Thus, the linear layer is not a static projection shared unchanged across different image sizes. Its parameters are adaptively generated according to the spatial metadata. The output spatial mapping is performed analogously using the target spatial metadata to produce the required output grid.

By adaptively mapping spatial–temporal–spectral dimensional configurations to a unified feature representation, DUM standardizes feature shapes while conditioning the mapping on the relationships encoded by the metadata. DUM also transforms this unified representation into task-specific output shapes. This enables a scenario-level jointly trained STSUN to support diverse input and output configurations, dense prediction tasks, and trained class subsets.

IV. EXPERIMENTAL SETTINGS AND RESULTS

To validate STSUN's adaptability to multiple input and output configurations, dense prediction tasks, and trained class subsets, we conducted experiments on seven datasets covering building and LULC scenarios, as summarized in Table I. These datasets span spatial resolutions from 0.075 to 10 m, input spectral configurations from 3 to 10 bands, and temporal sampling intervals from 1 month to 14 years. For each scenario, one STSUN model was jointly trained on the combined training sets of all included datasets and evaluated on their respective test sets; the experiments do not assess zero-shot transfer to an unseen task or class.

A. Datasets

1) Building Datasets: The WHU Building dataset [27] comprises an aerial photograph subset of 8189 images (512×512 pixels, 0.3 m resolution). Covering over 450 km^2 and containing more than 22 000 building instances, the data are partitioned into 4736 training, 1036 validation, and 2416 testing images following the original scheme.

The WHU-CD dataset [27] consists of bitemporal very high-resolution (VHR) aerial images from 2012 and 2016. In our unified formulation, the target is a two-step semantic sequence ($T_2 = 2$) of building/background maps, one for each date, and is therefore designated as SCD in Table I. The BC map used for evaluation is derived by comparing the two semantic predictions. The images are cropped into nonoverlapping 1024×1024 patches and allocated into training, validation, and test sets at a 7:1:2 ratio.

LEVIR-CD [28] provides 637 bitemporal VHR Google Earth image pairs (0.5 m/pixel, 1024×1024) spanning 5–14 years. It focuses on construction and demolition events, containing 31 333 BC instances (1 for change and 0 for no change). We adhered to its standard data partition.

The TSCD dataset [29] utilizes co-registered WorldView-2 satellite imagery (~0.5 m/pixel) from 2016, 2018, 2020, and 2022. Building footprints were densely labeled for each phase, and three sets of differential change labels (2016–2018, 2018–2020, and 2020–2022) were derived, uniformly cropped, and partitioned.

2) LULC Datasets: The LoveDA dataset [30] includes 5987 high-resolution optical images (0.3 m, 1024×1024) from urban and rural scenes across three Chinese cities. Covering seven land-cover classes, it presents challenges like multiscale objects and complex backgrounds. The dataset is split into 2522 training, 1669 validation, and 1796 testing images.

DynamicEarthNet [31] provides 55 Sentinel-2 image time series collected globally from 2018 to 2019. Each area of interest contains 24 monthly ground-truth segmentation maps. For SCD, STSUN predicts the full 24-step semantic sequence ($T_2 = 24$, $C_2 = 6$). For BCD, the label for each of the 23 adjacent month pairs is defined as changed when the two semantic labels differ and unchanged otherwise ($T_2 = 23$, $C_2 = 2$). The 1024×1024 multispectral (RGB and NIR) patches cover

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six LULC categories (impervious surface, agriculture, forest, wetlands, soil, and water), excluding the rare snow class.

MultiSenGE [32] is a multimodal benchmark containing 8157 image patches (256×256) from Sentinel-1 and Sentinel-2 sensors, covering eastern France (Alsace to Ardennes/Marne). Adopting the CSM configuration [56], this study employs the Sentinel-2 subset and a five-class land-cover nomenclature.

B. Benchmark Methods

To evaluate the effectiveness of the proposed model, we conducted comparative experiments with various benchmark methods on the building and LULC scenarios. The benchmark methods tested on the same dataset are based on the same splitting of the dataset and use the same data. Since STSUN is adaptable to multiple datasets, it was trained across all datasets within each scenario, whereas the baselines were limited to each dataset and trained on individual datasets. To disentangle the performance gains of increased training data from those of our unified approach, we introduced a strong baseline, $STSUN_{linear}$. This variant utilizes the same backbone as STSUN but replaces the unified dimension modules with dataset-specific linear layers. This allows us to evaluate the effectiveness of constructing a unified feature space via metadata and hypernetworks, compared to a hard-coded approach that uses multiple linear layers to accommodate diverse data–task–category configurations.

On the four building datasets, the compared CNN-based models include SegNet [42], U-Net [43], PSPNet [33], HRNet [34], MA-FCN [44], DeepLabv3+ [35], ResUNet-a [36], MAP-Net [37], SIINet [11], FC-EF [52], FC-Siam-Diff [52], FC-Siam-Conc [52], STANet [28], SNUNet [46], DDCNN [47], DSIFN [45], USSFCNet [53], and SEIFNet [55]; the compared transformer-based models include SegFormer [38], ChangeFormer [48], and A2Net [54]; and the CNN–transformer hybrid models include BDTNet [57], TransUNet [39], CMTFNet [40], CaSaFormer [41], BIT [9], MTCNet [49], MSCANet [50], AMTNet-50 [51], ContrastCOUD [29], and TS-COUD [29].

On the three LULC datasets, the compared CNN-based models include U-Net [43], HRNet [34], DeepLabv3+ [35], A2-FPN [58], and CAC [59]; the compared transformer-based models include SETR [60], SegFormer [38], Segmenter [61], and TSViT [62]; and the CNN–transformer hybrid models include UNetFormer [63], CMTFNet [40], SFFNet [64], SparseFormer [65], XNet [66], AFENet [67], UTAE [68], A2Net [54], CSM [56], SCanNet [69], and TSSCD [70].

C. Implementation Details

1) Data Augmentation: To validate the proposed methods, we adopted a minimalistic yet effective data augmentation strategy, deliberately refraining from complex augmentation schemes. In particular, the employed transformations were limited to horizontal/vertical flipping (probability = 0.5) and transposition (probability = 0.5).

2) Training and Inference: We implemented the STSUN model in PyTorch [71] using a single NVIDIA A100 GPU (80 GB). All models were initialized with the default PyTorch initialization. The network was optimized using a combined binary cross-entropy and Dice loss. We adopted the AdamW optimizer [72] with an initial learning rate of 0.0001 and a weight decay of 0.001. A learning-rate scheduler reduced the learning rate by a factor of 0.1 if the mean $F1$-score on the validation set did not improve for five consecutive epochs. Models were trained for 100 epochs, with batch sizes of 16 for the four building datasets and four for the three LULC datasets because of their resolution differences. The checkpoint with the highest validation mean $F1$-score was retained for testing.

3) Evaluation Metrics: Model performance was evaluated using overall accuracy (OA), precision (P), recall (R), $F1$-score, and intersection over union (IoU). For multitemporal and multiclass tasks, we report average $F1$-score (AF) and mean IoU (mIoU). Following the DynamicEarthNet protocol [31], we also report SC segmentation (SCS), BC, and SC scores to assess change localization and semantic classification jointly. For each metric, the best and second-best scores are highlighted in bold and underlined, respectively.

TABLE II
COMPARISON ON THE WHU DATASET

Methods	P (%)	R (%)	F1 (%)	IoU (%)
PSPNet [33]	93.19	94.21	93.70	88.14
HRNet [34]	91.69	92.85	92.27	85.64
DeepLabv3+ [35]	94.31	94.53	94.42	89.43
ResUNet-a [36]	94.49	94.71	<u>94.60</u>	<u>89.75</u>
MAP-Net [37]	93.99	94.82	94.40	89.40
SegFormer [38]	<u>94.72</u>	94.42	94.57	89.70
TransUNet [39]	94.05	93.07	93.56	87.89
CMTFNet [40]	90.12	**95.21**	92.59	86.21
CaSaFormer [41]	94.70	91.91	93.28	87.42
$STSUN_{linear}$	94.33	93.63	93.98	88.64
STSUN	**95.71**	<u>94.87</u>	**95.29**	**91.00**

D. Overall Comparison on the Building Dense Prediction Scenario

To validate the data–task–category unification capability of the proposed STSUN, we evaluated its performance across four diverse building-related datasets: WHU, WHU-CD, LEVIR-CD, and TSCD. These datasets exhibit high heterogeneity, encompassing distinct spatial resolutions (0.075–0.5 m), image sizes (256×256 and 1024×1024), and temporal configurations (single-phase, bitemporal, and multiphase sequences spanning up to 14 years). Furthermore, they represent three fundamental dense prediction tasks: semantic segmentation, BC detection, and SC detection. Instead of relying on task-specific architectural modifications, a unified STSUN framework was employed to handle these diverse inputs and outputs.

Quantitative comparisons against SOTA approaches are summarized in Tables II–V. The results demonstrate that STSUN consistently achieves competitive performance across all evaluated data-task configurations. In particular, STSUN attains the highest $F1$-scores and IoU metrics on every dataset, reaching 95.29%/91.00% on WHU, 92.13%/85.40% on

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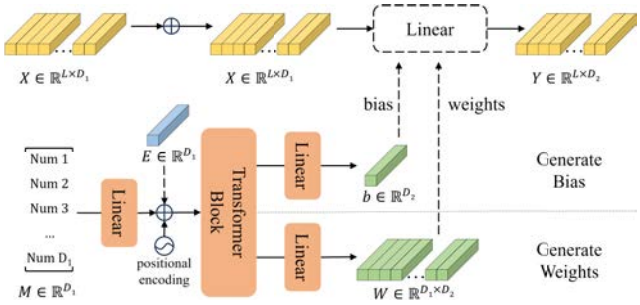


Fig. 4. Illustration of the DUM.

TABLE III
COMPARISON ON THE WHU-CD DATASET

Methods	P (%)	R (%)	F1 (%)	IoU (%)
SegNet [42]	85.20	86.21	85.70	74.98
DeepLabv3+ [35]	89.24	90.91	90.07	81.93
U-Net [43]	83.19	84.02	83.60	71.83
PSPNet [33]	84.85	82.09	83.45	71.60
HRNet [34]	86.77	85.92	86.34	75.97
MA-FCN [44]	86.10	89.92	87.97	78.52
SegFormer [38]	90.45	88.93	89.68	81.30
TransUNet [39]	**93.82**	89.33	91.52	84.37
CaSaFormer [41]	93.17	89.73	91.42	84.20
$STSUN_{linear}$	91.34	90.12	90.73	83.03
STSUN	93.07	**91.20**	**92.13**	**85.40**

TABLE IV
COMPARISON ON LEVIR-CD DATASET

Methods	P (%)	R (%)	F1 (%)	IoU (%)
DSIFN [45]	**94.02**	82.93	88.13	78.77
STANet [28]	83.81	**91.00**	87.26	77.39
SNUNet [46]	89.18	87.17	88.16	78.83
DDCNN [47]	91.85	88.69	90.24	82.22
BIT [9]	89.24	89.37	89.30	80.68
ChangeFormer [48]	92.05	88.80	90.40	82.47
MTCNet [49]	90.87	89.62	90.24	82.22
MSCANet [50]	91.30	88.56	89.91	81.66
AMTNet-50 [51]	91.82	89.71	90.76	83.08
$STSUN_{linear}$	92.08	88.86	90.44	82.55
STSUN	93.17	90.07	**91.59**	**84.49**

WHU-CD, 91.59%/84.49% on LEVIR-CD, and 66.48%/49.79% on TSCD. Notably, STSUN outperforms highly specialized, single-task architectures, such as SegFormer for semantic segmentation and ChangeFormer for change detection. This suggests that our STSUN not only affords structural flexibility but also facilitates the learning of robust, generalized spatial–temporal–spectral representations, mitigating the limitations associated with rigid, task-specific models. Furthermore, STSUN significantly outperforms $STSUN_{linear}$, demonstrating that constructing a unified feature space via metadata and hypernetworks is more effective than hard-coding data-task configurations through standard linear layers.

The qualitative performance of STSUN is further illustrated in Fig. 5, which presents representative inference results across the four datasets. Visually, STSUN generates accurate and coherent prediction maps with clear building boundaries, even

TABLE V
COMPARISON ON THE TSCD DATASET

Methods	F1 (%)	IoU (%)	OA (%)
FC-EF [52]	53.39	37.86	97.03
FC-Siam-Conc [52]	41.51	28.05	96.60
FC-Siam-Diff [52]	39.65	26.83	96.48
SNUNet-CD [46]	63.22	47.22	97.61
USSFCNet [53]	55.68	39.80	97.30
A2Net [54]	53.16	37.19	97.14
SEIFNet [55]	60.37	44.01	97.41
Contrast-COUD [29]	64.35	48.45	97.72
TS-COUD [29]	65.24	49.33	97.90
$STSUN_{linear}$	64.92	48.06	97.53
STSUN	**66.48**	**49.79**	**98.05**

in the presence of challenging multitemporal illumination differences and varying spatial scales. As indicated by the minimal false positives (red areas) and false negatives (blue areas), the model is robust to irrelevant background interference and temporal variations. For instance, in the single-phase semantic segmentation task [see Fig. 5(a)], STSUN effectively captures complex building footprints and shapes. Concurrently, in complex change detection scenarios involving long-term spans or multiple phases [see Fig. 5(c) and (d)], the temporal unification mechanism enables the model to identify actual construction and demolition events while suppressing pseudochanges.

E. Overall Comparison on the LULC Dense Prediction Scenario

The efficacy of the proposed STSUN is evaluated across three heterogeneous LULC datasets: LoveDA, DynamicEarthNet, and MultiSenGE. These datasets vary in modality, dense prediction task, and semantic category set. The input configurations range from single-temporal, high-resolution (0.3 m) optical images of size 1024 × 1024 in LoveDA to lower resolution (10 m) multispectral images of size 256 × 256 in MultiSenGE and 24-month image time series in DynamicEarthNet. The target definitions span semantic segmentation, BC detection, and SC detection, while the label spaces contain between two and seven classes.

The quantitative results are presented in Tables VI–VIII. Instead of requiring customized architectures or specific retraining for each scenario, STSUN accommodates the diverse input–output configurations and consistently outperforms specialized baseline methods. For instance, it achieves the highest mIoU on both the LoveDA (55.13%) and MultiSenGE (73.9%) datasets for semantic segmentation. On DynamicEarthNet, STSUN achieves the highest SCS (30.3%), BC (38.9%), and mIoU (55.7%) among the listed methods. Furthermore, STSUN consistently maintains its superiority over $STSUN_{linear}$. These comprehensive improvements underscore the effectiveness of our DUS and DUM. By overcoming the constraints of rigid data–task–category settings, STSUN not only jointly models the intrinsic correlations among varied spatial resolutions, temporal lengths, and spectral bands,

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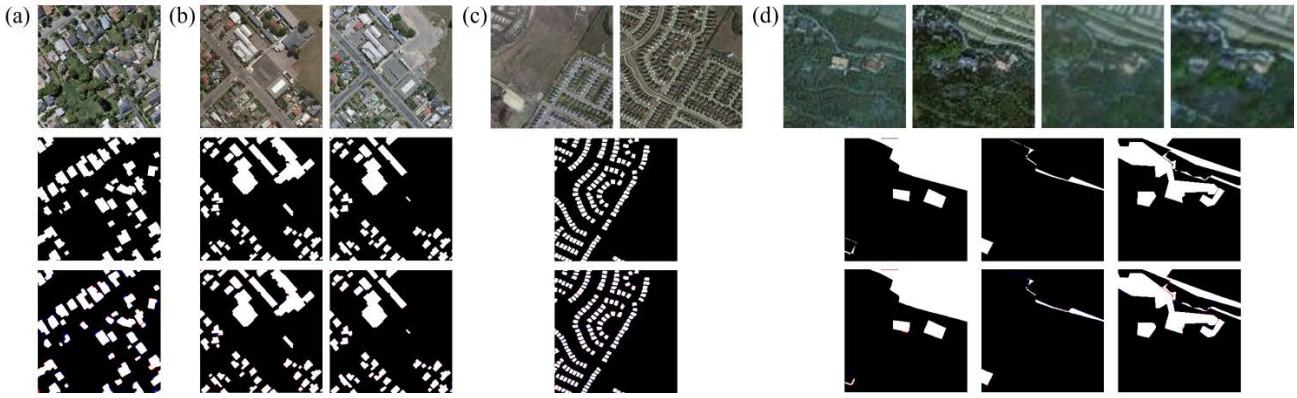


Fig. 5. Sample inference results on building-scene datasets. The first, second, and third rows show input images, ground truth, and predictions, respectively. Red areas denote false positives, and blue areas denote false negatives. (a) Semantic segmentation on WHU. (b) SC detection on WHU-CD. (c) BC detection on LEVIR-CD. (d) BC detection on TSCD.

TABLE VI
COMPARISON ON THE LOVEDA DATASET

Method	IoU per category (%)							mIoU (%)
	Background	Building	Road	Water	Barren	Forest	Agricultural	
U-Net [43]	43.38	53.53	53.52	76.03	17.75	44.74	58.33	49.61
DeepLabv3+ [35]	45.06	56.28	56.37	79.31	16.43	43.85	62.32	51.38
A^2-FPN [58]	44.84	57.25	55.68	78.73	20.79	46.10	62.68	52.29
UNetFormer [63]	44.59	58.82	55.36	79.92	19.10	46.15	61.31	52.18
CMTFNet [40]	46.20	58.57	56.06	79.60	19.88	46.60	63.83	52.96
SFFNet [64]	47.40	59.13	58.16	81.43	21.05	47.74	65.32	54.32
SparseFormer [65]	47.12	58.77	58.03	80.78	20.42	48.32	64.74	54.03
XNet [66]	46.35	59.05	58.96	81.37	21.28	48.19	65.89	54.44
AFENet [67]	47.43	59.22	59.17	81.55	21.36	48.68	66.40	54.83
$STSUN_{linear}$	46.73	58.92	58.84	81.17	21.02	48.35	66.07	54.44
STSUN	**48.02**	**59.49**	**59.29**	**81.68**	**21.79**	**48.82**	**66.85**	**55.13**

TABLE VII
COMPARISON ON THE MULTISENGE DATASET

Method	IoU per category (%)					mIoU (%)	OA (%)
	Build-Up	Cropland	Forest	Grassland	Water		
U-Net [43]	61.0	81.3	82.7	54.1	39.7	63.8	85.1
HRNet [34]	65.6	85.8	85.4	59.7	48.8	69.1	87.8
SETR [60]	50.7	76.6	77.6	47.8	31.9	56.9	81.1
Segmenter [61]	43.9	75.0	78.4	47.6	30.9	55.2	80.7
SegFormer [38]	59.7	82.2	82.8	55.6	39.4	63.9	85.3
CSM [56]	68.3	87.3	86.4	62.3	52.2	71.3	88.9
$STSUN_{linear}$	70.3	88.1	86.3	65.3	53.1	72.6	89.1
STSUN	**71.4**	**89.6**	**87.8**	**66.2**	**54.7**	**73.9**	**90.3**

but also exploits intertask complementarity and latent class relationships to enhance predictive accuracy.

These quantitative results are further corroborated by the qualitative visualizations depicted in Fig. 6. Despite the inherent challenges unique to each dataset, such as the multiscale objects and complex urban-rural backgrounds in LoveDA [see Fig. 6(a)], or the significant phenological and temporal variations inherent to the DynamicEarthNet sequence [see Fig. 6(b) and (c)], STSUN generates accurate and spatially coherent prediction maps.

F. Effect of Unifying Dense Prediction Tasks

To evaluate the capacity of STSUN to address rigid data and task constraints, we conduct a controlled comparison across

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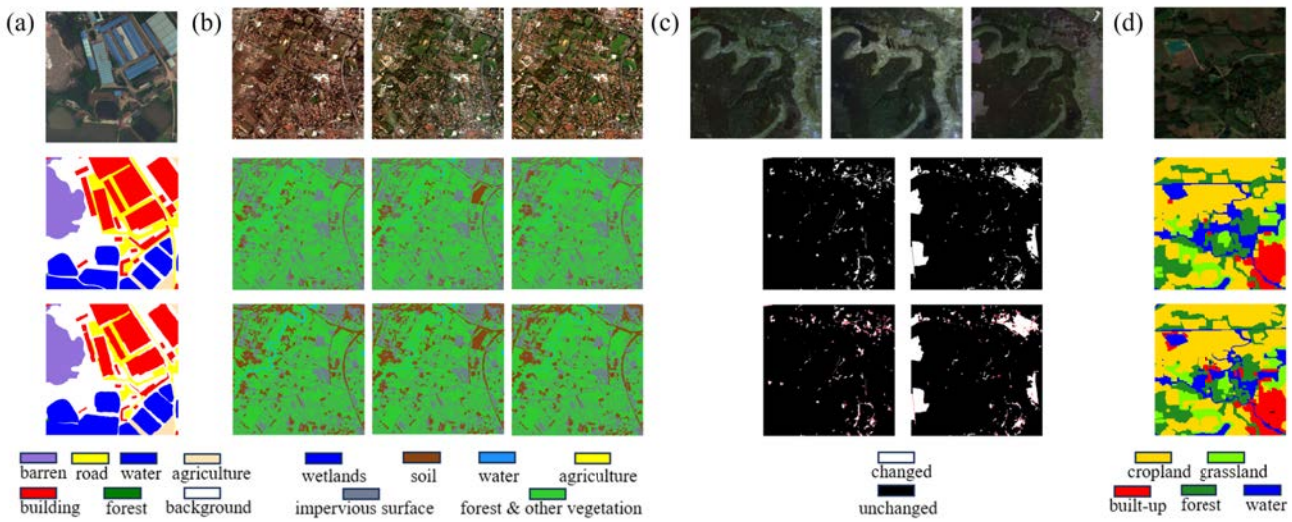


Fig. 6. Sample inference results on LULC-scene datasets. Within each panel, the rows show input images, ground truth, and predictions. (a) Semantic segmentation on LoveDA. (b) SC detection on DynamicEarthNet. (c) BC detection on DynamicEarthNet; red and blue denote false positives and false negatives, respectively. (d) Semantic segmentation on MultiSenGE.

TABLE VIII
COMPARISON ON THE DYNAMICEARTHNET DATASET

Methods	SCS ↑	BC ↑	SC ↑	mIoU (%) ↑
CAC [59]	17.7	10.7	**24.7**	37.9
U-Net [43]	17.3	10.1	24.4	37.6
TSViT [62]	23.0	34.1	11.8	50.5
UTAE [68]	25.9	38.0	13.8	53.7
A2Net [54]	22.2	32.9	11.5	47.2
SCanNet [69]	24.8	35.8	13.9	53.0
TSSCD [70]	12.0	19.4	4.7	33.9
$STSUN_{linear}$	25.1	37.1	13.1	53.2
STSUN	**30.3**	**38.9**	21.8	**55.7**

TABLE IX
COMPARISON OVER TASK UNIFICATION ON FOUR BUILDING DATASETS

Dataset	Methods	P (%)	R (%)	F1 (%)	IoU (%)
WHU	$STSUN_{single}$	95.24	94.30	94.77	90.06
	$STSUN_{linear}$	94.33	93.63	93.98	88.64
	$STSUN_{unified}$	**95.71**	**94.87**	**95.29**	**91.00**
WHU-CD	$STSUN_{single}$	92.22	90.89	91.55	84.42
	$STSUN_{linear}$	91.34	90.12	90.73	83.03
	$STSUN_{unified}$	**93.07**	**91.20**	**92.13**	**85.40**
LEVIR-CD	$STSUN_{single}$	92.94	89.63	91.25	83.92
	$STSUN_{linear}$	92.08	88.86	90.44	82.55
	$STSUN_{unified}$	**93.17**	**90.07**	**91.59**	**84.49**
TSCD	$STSUN_{single}$	66.91	65.08	65.98	49.23
	$STSUN_{linear}$	65.58	64.27	64.92	48.06
	$STSUN_{unified}$	**67.32**	**65.67**	**66.48**	**49.79**

four building datasets. These datasets contain varying patch sizes (e.g., 256×256 in TSCD and 1024×1024 in WHU-CD) and distinct dense prediction tasks: semantic segmentation in WHU, SC detection in WHU-CD, and BC detection in LEVIR-CD and TSCD. They also differ substantially in spatial resolution, temporal length, input configuration, and output structure, such that their joint training introduces considerable cross-dataset heterogeneity rather than merely increasing the number of training samples. We compare $STSUN_{single}$, which retains the complete model architecture but is trained independently on each dataset, and $STSUN_{linear}$, jointly trained across all four datasets using hard-coded dataset-specific linear layers, with $STSUN_{unified}$, which uses metadata-conditioned dimensional unification under the same joint-training data.

As detailed in Table IX, $STSUN_{unified}$ yields higher scores than $STSUN_{single}$ and the data-matched $STSUN_{linear}$ across all four datasets. Importantly, $STSUN_{single}$ consistently outperforms $STSUN_{linear}$. Although $STSUN_{linear}$ is exposed to more datasets and prediction tasks, its fixed, dataset-specific mappings fail to resolve the heterogeneity among their data distributions, temporal configurations, task definitions, and output spaces. This result demonstrates that simply combining more data and tasks does not guarantee performance improvements; unresolved cross-dataset heterogeneity can instead lead to negative transfer and degrade performance relative to independent training.

On the contrary, $STSUN_{unified}$ consistently outperforms $STSUN_{single}$, showing that the proposed dimensional unification mechanism enables the jointly trained model to effectively exploit complementary supervision from heterogeneous datasets rather than merely benefiting from a larger training set. Because $STSUN_{unified}$ and $STSUN_{linear}$ use the same backbone and the same joint-training data, their comparison more directly isolates the contribution of the DUMs. Dimensional metadata and hypernetworks generate adaptive mappings that construct a shared representation while retaining configuration-specific transformations.

Accordingly, the comparisons with $STSUN_{single}$ and $STSUN_{linear}$ provide the controlled evidence required to attribute the improvements to the proposed unification mechanism, whereas the comparisons with specialized methods trained on individual datasets evaluate the overall adaptability

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TABLE X
COMPARISON OVER CATEGORY UNIFICATION ON TWO LULC DATASETS

Model	LoveDA Dataset		DynamicEarthNet Dataset	
	AF↑	mIoU↑	SCS↑	mIoU↑
$STSUN_{single}$	74.66	55.02	30.0	55.3
$STSUN_{fixed}$	74.03	54.37	29.4	54.2
$STSUN_{flexible}$	**74.81**	**55.13**	**30.3**	**55.7**

and practical value of using a single model across multiple data–task–category settings.

G. Effect of Flexible Semantic Category Set

To demonstrate the efficacy of STSUN in handling varying semantic category sets, we designed a comparative experiment utilizing the LoveDA and DynamicEarthNet datasets within the LULC scenario. These datasets not only encompass different temporal scales but also feature distinct label spaces (seven urban/rural classes versus six global land-cover classes). We evaluated three configurations: 1) $STSUN_{single}$, trained independently to predict dataset-specific categories; 2) $STSUN_{fixed}$, a jointly trained model employing a static, ten-category output space corresponding to the union of both label sets; and 3) $STSUN_{flexible}$, our proposed approach that utilizes the DUM to dynamically align predictions with the relevant semantic embedding subset for each scene.

As shown in Table X, $STSUN_{flexible}$ outperforms the $STSUN_{fixed}$ approach on both datasets and achieves performance comparable to the individually trained $STSUN_{single}$ models. The degraded performance of $STSUN_{fixed}$ highlights a limitation of fixed category sets: applying a static, unified output space across varying scenarios introduces semantic ambiguity, as the model may be confused by categories that are absent from a given scene's taxonomy. Conversely, $STSUN_{flexible}$ successfully mitigates this interference by adaptively tailoring its predictive output to the target scene's specific semantic context. Ultimately, $STSUN_{flexible}$ maintains high accuracy while offering improved deployment efficiency. A single, unified model can generalize across scenes with highly variable semantic categories, reducing the need for dataset-specific retraining or fine-tuning.

H. Ablation Study

To evaluate the effectiveness of STSUN's DUS, the LGWA module, and metadata-guided adaptive mapping, we conducted ablation studies on four datasets: WHU, WHU-CD, LEVIR-CD, and TSCD. The proposed strategy, referred to as decoupled unification, maintains the independence of the temporal dimension. For comparison, we introduced a baseline termed coupled unification, which directly unifies the spatial, temporal, and spectral dimensions at both the input and output stages. In addition, LGWA was replaced with standard global attention or window attention [73] to assess its contribution. To explicitly isolate the effect of metadata, we further constructed a metadata-free counterpart for each configuration by replacing all DUMs with dataset-specific linear layers while keeping the remaining architecture and joint-training setting unchanged.

Table XI presents the ablation results on the four datasets. Decoupled unification consistently outperforms coupled unification across all attention configurations and both metadata settings, while LGWA achieves better performance than standard global attention and window attention under both unification strategies. In particular, the complete configuration combining decoupled unification, LGWA, and metadata-guided mapping yields the best $F1$ and IoU scores across all four datasets, demonstrating the complementary contributions of these components. Compared with coupled unification using LGWA and metadata, the complete configuration improves the $F1$ score by 1.54, 1.72, 1.66, and 3.26 percentage points on WHU, WHU-CD, LEVIR-CD, and TSCD, respectively. The effectiveness of LGWA can be attributed to its comprehensive receptive field. Integrating local attention across multiple window sizes with global context enables the model to capture both fine-grained geometric details and broader spatial context.

The advantage of decoupled unification is consistent with established findings on early and feature-level fusion [7], [10], [52]. Coupled unification combines multitemporal observations at the input stage, resembling early pixel-level fusion, which may entangle task-relevant temporal differences with phase-specific background variations. Moreover, it restores the required output temporal length only at the final prediction stage, limiting the model's capacity for task adaptation. On the contrary, decoupled unification first processes each temporal image independently through the shared spatial–spectral mapping modules and encoder, and then performs metadata-conditioned feature-level fusion through the TUM. This preserves phase-specific spatial–spectral representations before temporal integration and reduces interference from irrelevant information. The fused features are subsequently mapped to the requested output temporal length and further refined by the decoder, enabling more effective adaptation to different dense prediction tasks.

Finally, the metadata ablation verifies the effectiveness of metadata-guided adaptive mapping. Across all unification-strategy and attention configurations, replacing the DUMs with dataset-specific linear layers consistently reduces $F1$ and IoU on all four datasets. Since the remaining architecture and joint-training setting are unchanged, these results demonstrate that metadata-conditioned hypernetworks provide more effective configuration-adaptive mappings than hard-coded dataset-specific linear projections.

V. DISCUSSION

The proposed framework formulates remote sensing dense prediction as a unified spatial–temporal–spectral mapping problem rather than a collection of task-specific architectures. By relaxing rigid input–output constraints through the DUM and a decoupled unification strategy, one scenario-level model can share representations across heterogeneous supervised datasets while retaining metadata-conditioned mappings for each configuration.

The controlled STSUN-family comparisons suggest that metadata-conditioned unification is preferable to dataset-specific linear mappings under the same joint-training data. Joint training also exposes the network to a broader range of

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TABLE XI
ABLATION STUDY OF THE UNIFICATION STRATEGY, ATTENTION TYPE, AND METADATA-GUIDED ADAPTIVE MAPPING ON THE WHU, WHU-CD, LEVIR-CD, AND TSCD DATASETS

Unification Strategy	Attention Type	Metadata	WHU		WHU-CD		LEVIR-CD		TSCD	
			F1 (%)	IoU (%)	F1 (%)	IoU (%)	F1 (%)	IoU (%)	F1 (%)	IoU (%)
Coupled Unification	Global	×	91.97	85.13	88.58	79.50	88.22	78.92	61.50	44.40
		✓	93.11	87.11	89.79	81.47	89.30	80.67	62.87	45.85
	Window	×	91.46	84.26	88.19	78.87	87.86	78.35	61.22	44.11
		✓	92.68	86.36	89.37	80.78	88.91	80.03	62.63	45.59
	LGWA	×	92.50	86.05	89.10	80.34	88.81	79.87	61.76	44.68
		✓	93.75	88.24	90.41	82.50	89.93	81.70	63.22	46.22
Decoupled Unification	Global	×	93.53	87.85	90.21	82.17	89.94	81.72	64.47	47.57
		✓	94.71	89.95	91.50	84.33	91.04	83.55	65.91	49.15
	Window	×	93.19	87.25	89.89	81.64	89.63	81.21	64.19	47.26
		✓	94.43	89.45	91.22	83.86	90.76	83.08	65.68	48.90
	LGWA	×	93.98	88.64	90.73	83.03	90.44	82.55	64.92	48.06
		✓	**95.29**	**91.00**	**92.13**	**85.41**	**91.59**	**84.48**	**66.48**	**49.79**

spatial–temporal–spectral patterns, although the current experiments do not fully separate the effects of data volume and task complementarity. For semantic category unification, STSUN addresses a closed-set multitaxonomy setting rather than open vocabulary recognition. The class-embedding library avoids applying irrelevant union classes to every sample by selecting the category subset encoded in M_{C_2}, from which the DUM generates the corresponding class-dependent output mapping. As demonstrated by the category-unification experiment in Section IV-G, the flexible formulation consistently outperforms the fixed union-category formulation on both LoveDA and DynamicEarthNet, indicating that dynamically selecting the relevant category subset alleviates semantic interference from classes absent from a particular dataset. Nevertheless, this flexibility remains limited to supervised class identities represented by learned embeddings and does not extend to entirely unseen categories.

STSUN is conceptually distinct from existing remote sensing foundation models in both architecture and application paradigm. Foundation models typically aim to learn broadly transferable representations through large-scale pretraining and are subsequently adapted to downstream tasks, often with task- or category-specific prediction heads. On the contrary, STSUN is a supervised end-to-end dense prediction framework that uses metadata-conditioned hypernetworks, task embeddings, and category embeddings to unify heterogeneous input and output configurations within a shared encoder–decoder. Its primary objective is therefore data–task–category unification rather than general-purpose representation pretraining. Nevertheless, the two directions are complementary, as STSUN's unified input–output interface could be integrated with pretrained foundation-model backbones to support heterogeneous remote sensing data and joint adaptation across multiple dense prediction tasks and label spaces.

While the proposed STSUN successfully harmonizes heterogeneous remote sensing data, its current instantiation presents limitations that outline distinct avenues for future research.

1) *Reliance on Explicit Metadata:* The flexibility of STSUN depends on metadata provided during inference and therefore assumes prior knowledge of the requested analytical objective. Future work will investigate whether task configurations can be inferred reliably from contextual data cues while preserving user control over the intended output.
2) *Closed-Set Category Constraints:* Although STSUN supports flexible category sets, it currently relies on a predefined, trainable library of class embeddings updated solely via visual supervision. Accordingly, the current implementation can select only among known category subsets and should not be interpreted as an OVSS model; it cannot recognize a category for which no embedding has been learned. To achieve authentic open vocabulary recognition, future iterations will integrate pretrained text encoders to replace static embeddings with dynamic, language-aligned representations.
3) *Generalization Under a Supervised Paradigm:* Dimensional unification does not by itself guarantee semantic generalization across unseen scenes, sensors, tasks, or classes. The present evidence is limited to supervised joint training within two scenarios, and the external-baseline comparisons are not data-budget matched. Extending the architecture to self-supervised or vision–language pretraining and evaluating transfer under controlled protocols are future research directions rather than demonstrated capabilities.

VI. CONCLUSION

We propose STSUN, an architecture that addresses rigid data–task–category configurations in remote sensing dense prediction. Metadata-conditioned hypernetworks map variable

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spatial, temporal, and spectral inputs and outputs to and from a shared representation. Within each scenario, one jointly trained model integrates semantic segmentation, BC detection, and SC detection and selects among the class subsets represented during training. Experiments on seven datasets show competitive performance and consistent gains over the data-matched linear-unification variant. The present results support STSUN as a supervised unified dense-prediction architecture. They do not establish zero-shot unseen-class generalization or a remote sensing foundation model.

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